PRODUCT USER MANUAL

For Baltic Sea Ice Observations

SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011

Issue: 1.3

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.3	14/03/2025	All	Adaptation to new template, adding SID mosaic (EiS 11/2025)	Juha Karvonen	
1.3	20/05/2025	All	Inclusion of format change to netcdf4 (EIS 11/2025)	Juha Karvonen	C. Wettre

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
MDS	Marine Data Store
RMS	Root mean square
PC	Production Centre
PU	Production Unit
SIC	Sea ice concentration
SIT	Sea ice thickness
CF	Climate and forecast metadata convention
IC	Ice chart
ITC	Ice thickness chart
ASIC	Automated SIC
ASIT	Automated SIT

SIE	Sea ice extent
SID	Sea ice drift
IDC	Ice drift chart
CSK	Cosmo-SkyMED, Itlaian X-band SAR (constellation)
TSX	TerraSAR-X, German X band SAR
PAZ	Spanish X-band SAR
RCM	Radarsat constellation mission, Canadian C-band SAR constellation
RS-2	Radarsat-2, Canadian C-band SAR
S-1	Sentinel-1. C-band SAR constellation of ESA
FMI	Finnish Meteorological Institute
SMHI	Swedish Meteorologicaln and Hydrological Institute

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

INTRODUCTION

a) Summary

This user manual describes the regional Baltic sea ice product provided by the Copernicus Marine Sea Ice Thematic Ensemble Centre (SITAC). The product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011 datasets described in this document are based on automated interpretation of earth bservation data, mainly from syntehtic aperture radar (SAR). These product datasets are produced by FMI and disseminated through the central Copernicus Marine dissemination unit. They are high resolution regional sea ice near-real-time products over the Baltic Sea region. The data are freely available under the Copernicus Marine data license.

This product includes eight datasets.

SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011 datasets:

- **FMI-BAL-SEAICE_THICK-SAR-NRT-OBS**: gridded ice thickness based on the digitized FMI ice charts refined by SAR. It is produced after reception of a SAR image either from Sentinel 1 (A and B), Radarsat-2 or Radarsat Constellation Mission (RCM). The nominal resolution is about 500m.
- **FMI-BAL-SEAICE_DRIFT-SAR-NRT-OBS**: gridded ice motion based on the SAR data. Production of two overlapping SAR images with a time difference less than or equal to two days. It is produced twice a day if suitable SAR image pairs after the previous production have appeared. SAR images are either from Sentinel 1 or Radarsat-2. Nominal resolution is about 800m.
- **cmeme_obs-si_bal_phy-sit_nrt_l4_P1D_m**: gridded ice thickness mosaic covering the Baltic Sea based on the digitized FMI ice charts refined by SAR. It is produced after reception of a SAR image Sentinel-1, Radarsat-2, RCM or any X-band SAR. The nominal resolution is about 500m.
- **cmems_sat-si_bal_conc_nrt_500m_d** : gridded SIC daily mosaic over the Baltic Sea based on SENTINEL-1 EW and RADARSAT-2 SCW dual-polarized (HH/HV) imagery.
- **cmems_sat-si_bal_thick_nrt_500m_hi** : SIT based on Sentinel-1 IW (VV/VH polarization) imagery, computed for each IW image separately and delivered after reception of each Sentinel-1 IW image over the Baltic Sea.
- **cmems_obs-si_bal_phy-sit_nrt_x-500m-l4_P1D** : SIT based on X-band SAR (HH polarization) imagery from CSK, TSX and PAZ, computed for each image separately and delivered after reception of each X-band image over the Baltic Sea.
- **cmems_sat-si_bal_sie_nrt_500m_d** : daily sea ice extent derived from the cmems_sat-si_bal_conc_nrt_500m_d dataset.
- **cmems_obs-si_bal_phy-sid_nrt_l4_P1D-m** : daily sea ice drift mosaic. Because SID for different image pairs has varying time differences, most recent drift speed at each grid point is provided in the SID mosaic dataset.

DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

The product datasets cover the Baltic Sea. The datasets based on single satellite images cover parts, corresponding to each satellite image area, of the Baltic Sea area and the mosaic datasets cover the whole area. The area is indicated in Figure 1.

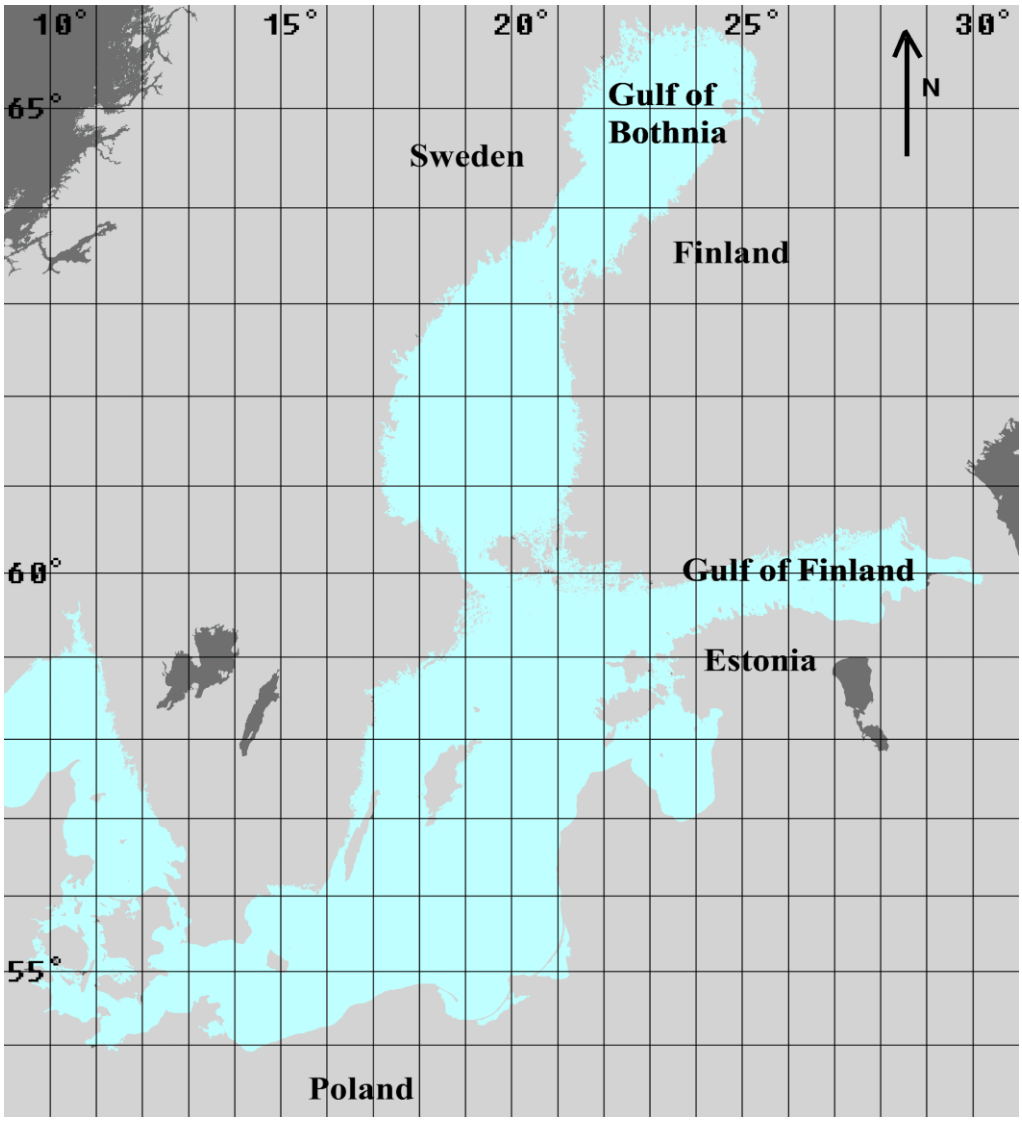

Figure 1. The Baltic Sea area (light blue color) is approximately between 53 and 66 degrees of Northern latitude and 9 and 31 degrees of Eastern longitude.

Product	Baltic Sea Ice : SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011
Geographical coverage	Baltic Sea : 9° E - 31° E; 53° N - 66° N, coverage of the SAR image area for single NRT data, whole area for the mosaic products
Variables	• Sea Ice Concentration • Sea Ice Thickness • Sea Ice Drift • Sea Ice Extent
Product Type	Near Real Time - Observations
Available time series	-2Y
Temporal resolution	Varying (for observations derived from satellite image scenes soon after acquisition) / Daily mosaics
Update Frequency	Varying (for observations derived from satellite image scenes soon after acquisition) / Daily mosaics
Target delivery time	4 hours after SAR acquisition or daily (mosaics)
Resolution	0.5 (SIC, SIT) / 0.8 km (SID)
Number of vertical levels	1
Format	NetCDF v. 3 or 4 CF1.4

b) Production System Description

Automatic Sea ice thickness chart (ITC)

An ITC is produced after receiving a SAR image either from Sentinel-1, Radarsat-2 and Radarsat Constellation Mission (RCM). The HH channel wide swath images (Sentinel-1 EW mode and Radarsat-2/RCM SCW mode images are used to generate the product. The product covers the sea ice area covered by the SAR image. The resolution of the product is 500 m.The product is based on the sea ice thickness history from the most recent IC available before the SAR. The SAR data is preprocessed by performing an incidence angle correction and speckle filtering. After this preprocessing a segmentation is performed, and segment wise image autocorrelation feature is computed for the image. This textural feature is used to locate the open water areas. From the IC ice thickness range is assigned to each SAR segment, i.e. a mapping of the SAR segments to IC segments is performed. This improves the spatial accuracy of the IC segments. After this an ice thickness value is assigned to each SAR segment. This thickness value is dependent on the ice thickness range adopted from the IC and on the mean SAR segment backscattering such that higher SAR backscattering corresponds to thicker ice. Finally, the open water areas are updated according to the SAR open water classification. Since December 2020 a new data set of ice thickness-based Sentinel-1 IW VV/VH dual-polarized mode was included. Inclusion of this data set improves the temporal resolution of the SIT products. The resolution and timeliness of the new data set are similar as for the EW/SCN mode data set. Since December 2023 a new data set providing SIT based on X-band HH-polarized SAR data has been included. The X-band SAR instruments currently used are CSK, TSX and PAZ. Inclusion of X-band SAR SIT further improves the temporal coverage.

PARAMETERS
Inputs: IC (ice thickness history), georeferenced Sentinel 1 WSM or Radarsat-2 SCW (level 1b data)
Resolution: 500m
Coverage: SAR area (SAR imagery over the Baltic Sea area)
Since: Dec 2010 (depending on the existence of the Baltic Sea ice cover)
Temporal resolution: Dependent on the received SAR data
Units: cm

For details on the algorithm, see Karvonen et al., 2003. Karvonen and Cheng, 2024

Automatic Sea ice Drift Chart (IDC)

An IDC is produced after receiving two SAR images over a same area in the Baltic Sea with a time gap of less than three days between the SAR images. Each received SAR image is studied after it has been received and if it has common areas with an earlier SAR image less than three days older, the product is computed. The images can currently be either from Sentinel-1 (EWS mode), Radarsat-2 or RCM (SCW mode). The drift estimates between pairs of images from different instruments (mixed pairs) are also computed. The product is based on computing phase correlation of pairwise data windows sampled from the two images in two resolutions. In the coarse resolution the large-scale motion is located and in the fine resolution these motions are refined. In the SAR areas without features suitable for computation of correlation the drift is not

computed. Such featureless areas include level ice areas and open water. The product gives estimates of the ice drift between the images (or mean velocities between the images), these are naturally just estimates of the integrated ice trajectories between the time instants corresponding to SAR acquisition times. The results are given in 800m resolution, which is half of the window size (16x16 pixels for a SAR image with a pixel size of 100m).

PARAMETERS
Inputs: Two georeferenced wide swath SAR images (level 1b) over the same area.
Resolution: 800m
Coverage: Common area of the two SAR images (in the Baltic Sea area)
Since: Dec 2010 (depending on the existence of the Baltic Sea ice cover)
Temporal resolution: Dependent on the received SAR data
Units: m

For details of the algorithm, see Karvonen 2012.

Automatic Sea ice thickness mosaic

An ITC is produced twice daily (8 UTC and 18 UTC) if new C- or X-band SAR HH-polarized data are available. This is done by overlaying the most recent ITC over the older one. The maximum ITC age in the mosaic has been restricted to one week, in the location where the SAR data is older, no thickness is given.
PARAMETERS
Inputs: derived from the single SAR SIT products
Resolution: 500m
Coverage: (66N,9E) - (53N,31E)
Since: March 2016 (depending on the existence of the Baltic Sea ice cover)
Temporal resolution: Provided daily if SAR data are available
Units: cm

Automatic Sea ice drift mosaic

An ITC is produced twice daily (9 UTC and 17 UTC) if new C-band SAR based SID data are available. This is done by overlaying the most recent SID over the older one. Bacause of different time spans between different image pairs, ice drift velocity is provided in the SID mosaic.
PARAMETERS
Inputs: derived from the single SAR SID data products
Resolution: 800m
Coverage: (66N,9E) – (53N,31E)
Since: Dec 2025 (depending on the existence of the Baltic Sea ice cover)
Temporal resolution: Provided daily if SID data are available
Units: mm/s

Automatic Sea ice concentration chart (ICC)

The daily ICC (mosaic) is produced daily (at 10 UTC) based on the most recent SAR data and the AMSR2 microwave radiometer data acquired during the past 24 hours. The most recent results is always overlaid at each location and thus the most recent

information available at each pixel location.
PARAMETERS
Inputs: Sentinel 1 WSM and Radarsat-2 SCW dual-polarizes SAR (level 1b data) and AMSR2 microwave radiometer brightness temperatures
Resolution: 500m
Coverage: Baltic Sea (66N,9E) - (53N,31E)
Since: July 2020
Units: %
For details of the algorithm, see Karvonen, 2017.

PARAMETERS
Inputs: C-band SAR HH/HV and AMSR2 microwave radiometer mosaic.
Resolution: 500m
Coverage: Baltic Sea (daily mosaic)
Since: Dec 2020
Temporal resolution: 1 day

Automatic Sea ice Extent (SIE)

The daily automated sea ice extent is derived from the automated sea ice concentration data set. It is provided daily after the automated SIC. It is given in 1e6km2 for the whole Baltic Sea.
Inputs: Sentinel-1, Radarsat-2 HH/HV SAR data and AMSR-2 microwave radiometer data.
Coverage: the whole Baltic Sea area.
Since: 01.12.2024
Units: 1e6km2

c) Dataset details

Automated sea ice thickness based on wide swath HH/HV-polarized C-band SAR data

Product: SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	FMI-BAL-SEAICE_THICK-SAR-NRT-OBS
Geographical coverage	Baltic Sea 9° E □ 31° E ; 53° N □ 66° N
Variables	Ice thickness (cm)
Analysis	SAR (Sentinel-1 EW, Radarsat-2 SCW dual-polarized HH/HV)
Forecast	No
Available time series	- 2 years - on going
Temporal Resolution	Depending on SAR data
Target delivery time	After each SAR reception, during Baltic Sea ice season
Delivery mechanism	MDS

Resolution	500 m
Number of vertical levels	1
Format	netCDF-3 CF1.4

Automatic Sea ice thickness chart (ITC) based on C-band SAR VV/VH data (S-1 IW mode)

Product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	cmems_sat-si_bal_thick_nrt_500m_hi
Geographical coverage	Baltic Sea 9° E □ 31° E ; 53° N □ 66° N
Variables	Ice thickness (cm)
Analysis	SAR (Sentinel-1 IW VV/VH mode)
Forecast	No
Available time series	From 15 December 2020 - on going
Temporal Resolution	Depending on SAR data
Target delivery time	After each SAR reception, during Baltic Sea ice season
Delivery mechanism	MDS
Resolution	500 m
Number of vertical levels	1
Format	netCDF-4 CF1.4

Automatic Sea ice thickness chart (ITC) based on X-band SAR HH data

Product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	cmems_obs-si_bal_phy-sit_nrt_x-500m-l4_P1D
Geographical coverage	Baltic Sea 9° E □ 30° E ; 53° N □ 66° N
Variables	Ice thickness (cm)
Analysis	X-band SAR (CSK/TSX/PAZ HH polarization)
Forecast	No
Available time series	From 1 December 20203- on going
Temporal Resolution	Depending on SAR data
Target delivery time	After each SAR reception, during Baltic Sea ice season
Delivery mechanism	MDS
Resolution	500m

Number of vertical levels	1
Format	netCDF-4 CF1.4

Automatic Sea ice Drift chart (IDC) based on multitemporal C-band SAR HH/HV data

Product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	FMI-BAL-SEAICE_DRIFT-SAR-NRT-OBS
Geographical coverage	Baltic Sea 9° E □ 30° E ; 53° N □ 66° N
Variables	Displacement in X and Y; ice drift
Analysis	No
Forecast	No
Available time series	From January 2011 - on going
Temporal resolution	Depending on SAR data
Target delivery time	Daily, during Baltic sea ice season
Delivery mechanism	MDS
Resolution	800m
Number of vertical levels	1
Format	netCDF-4 CF1.4

Automatic Sea ice Drift mosaic based on multitemporal C-band SAR HH/HV data

Product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	cmems_obs-si_bal_phy-sid_nrt_l4_P1D-m
Geographical coverage	Baltic Sea 9°E □ 30°E ; 53°N □ 66°N
Variables	Sea ice drift velocity in X and Y
Analysis	No
Forecast	No
Available time series	From Dec 2025 – on going
Temporal resolution	Depending on SAR data availability
Target delivery time	Daily, during Baltic sea ice season
Delivery mechanism	MDS
Resolution	800m
Number of vertical levels	1
Format	netCDF-4 CF1.4

Automatic Sea ice thickness chart mosaic (ITM)

Product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	cmems_obs-si_bal_phy-sit_nrt_l4_P1D_m
Geographical coverage	Baltic Sea 9° E □ 31° E ; 53° N □ 66° N
Variables	Ice thickness (cm)
Analysis	SAR
Forecast	No
Available time series	- 2 years - on going
Temporal resolution	Updated twice daily over the areas with new C- or X-band SAR data
Target delivery time	8 UTC and 18 UTC
Delivery mechanism	MDS
Horizontal resolution	500m
Number of vertical levels	1
Format	netCDF4 CF1.4

Automatic Sea ice concentration (ITC) based on C-band SAR HH/HV data and microwave radiometer (AMSR2) data

Product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	cmems_sat-si_bal_conc_nrt_500m_d
Geographical coverage	Baltic Sea 9° E □ 31° E ; 53° N □ 66° N
Variables	Ice concentration (%)
Analysis	SAR /MWR
Forecast	No
Available time series	From March 2020 - on going
Temporal resolution	Updated daily
Target delivery time	10 UTC
Delivery mechanism	MDS
Resolution	500m
Number of vertical levels	1
Format	netCDF-4 CF1.4

Sea Ice Extent (SIE) based on automatic SIC

Product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011	
Data set	cmems_obs-si_bal_phy-sie_nrt_500m_P1D-m
Geographical coverage	Baltic Sea 9° E □ 31° E ; 53° N □ 66° N
Product Lines	Sea Ice Extent (1e6km2)
Analysis	SAR / MWR
Forecast	No
Available time series	From December 2024
Temporal resolution	1 day
Target delivery time	Daily, during Baltic sea ice season
Delivery mechanism	MDS
Resolution	NA
Number of vertical levels	1
Format	netCDF-4 CF1.4

Variables by dataset:

SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_011 DATASETS	
DATASET	Variable's name in the NetCDF file and Unit Long_name CF Standard_name
FMI-BAL-SEAICE_THICK-SAR-NRT-OBS cmems_sat-si_bal_thick_nrt_500m_hi cmems_obs-si_bal_phy-sit_nrt_x-500m-l4_P1D	sea_ice_thickness [m] SAR (S-1, RS-2, RCM) based sea ice thickness sea_ice_thickness
FMI-BAL-SEAICE_DRIFT-SAR-NRT-OBS	sea_ice_x_displacement [m] sea ice drift U coordinate sea_ice_x_displacement
	sea_ice_y_displacement [m] sea ice drift V coordinate sea_ice_Y_displacement
	ice_drift_quality [%] sea ice drift quality
cmems_obs-si_bal_phy-sid_nrt_l4_P1D-m	sea_ice_x_velocity (mm/s) sea ice drift U coordinate sea_ice_x_velocity
	sea_ice_y_velocity (mm/s) sea ice drift U coordinate sea_ice_y_velocity

cmems_obs-si_bal_phy-sit_nrt_l4_P1D-m (ice thickness mosaic)	ice_thickness [cm] FMI ice chart sea ice thickness in cm sea_ice_thickness
	Data_age [d] SAR data age in days for each grid point
cmems_sat-si_bal_conc_nrt_500m_d	Sea ice fraction [%] Baltic Sea ice concentration sea_ice_area_fraction
cmems_obs-si_bal_phy-sie_nrt_500m_P1D-m	Sea ice extent (1e6km2) Baltic Sea ice extent sea_ice_extent

d) Additional information on some parameters

N/A

e) Processing information

Product disseminating schedule :

Type of product	Observation time	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
NRT EO observation	Dependent on EO data acquisition	N/A	4 hours after EO data acquisition	According to EO data availability
Daily mosaic datasets (three datasets : cmems_obs-si_bal_physit_nrt_l4_P1D-m and cmems_satsi_bal_conc_nrt_500m_d , cmems_obs-si_bal_physid_nrt_l4_P1D-m)	12 UTC	1 day	14-15 UTC	Every day

f) Catalogue updating

After dataset updates (irregular).

g) Time averaging

N/A

h) Product use and impact

The major use for the data are winter navigation and other offshore and coastal operations, e.g. off-share and coastal construction. Can also be used e.g. in numerical weather and ice model data assimilation and evaluation.

FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

a) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
FMI-BAL-SEAICE_THICK-SAR-NRT-OBS	ice_thickness_baltic_YYYYMMDDHHMMSS.nc	0.002-0.007 (x up to 20 files daily)
cmems_obs-si_bal_phy-sit_nrt_l4_P1D-m	ice_thick_mosaic_baltic_YYYYMMDDHHMMSS_YYYYMMDDHHMMSS.nc	0.029 daily
FMI-BAL-SEAICE_DRIFT-SAR-NRT-OBS	ice_drift_baltic_YYYYMMDDHHMMSS_YYYYMMDDHHMMSS.nc	0.00025 (x up to 10 files daily)
cmems_sat-si_bal_conc_nrt_500m_d	ice_conc_mosaic_baltic_YYYYMMDDHHMMSS_YYYYMMDDHHMMSS.nc	0.029 daily
cmems_sat-si_bal_thick_nrt_500m_hi	ice_thickness_baltic_IW_YYYYMMDDHHMMSS.nc	0.001 (x up to 8 files daily)
cmems_sat-si_bal_sie_nrt_500m_d	sie_baltic_eo_YYYYMMDD.nc	0.0000013
cmems_obs-si_bal_phy-sit_nrt_x-500m-l4_P1D	ice_thickness_baltic_X_YYYYMMDDHHMMSS.nc	0.001 (x 1-5 files daily)
cmems_obs-si_bal_phy-sid_nrt_l4_P1D-m	ice_drift_mosaic_baltic_YYYYMMDDHHMMSS.nc	0.04 daily

b)STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SEAICE-QUID-011-004-011-019.pdf

J. Karvonen, M. Simila, I. Heiler, Ice Thickness Estimation Using SAR Data and Ice Thickness History, Proceedings of the IEEE International Geoscience and Remote Sensing Symposium 2003 (IGARSS 03), v. I, pp. 74-76, Toulouse, France, 2003.

J. Karvonen. Operational SAR-based sea ice drift monitoring over the Baltic Sea, Ocean Science, v. 8, pp. 473-483, (http://www.ocean-sci.net/8/473/2012/os-8-473-2012.html) 2012

J. Karvonen, Baltic Sea Ice Concentration Estimation Using SENTINEL-1 SAR and AMSR2 Microwave Radiometer Data, IEEE Transactions on Geoscience and Remote Sensing, pp. 2871 - 2883, v. 55, n. 5, 2017.

J. Karvonen, B. Cheng, Baltic sea ice thickness estimation based on X-band SAR data and background information. *Annals of Glaciology*. 2024;65:e23. doi:10.1017/aog.2024.24

ANNEX

These NetCDfF headers are examples of the operative datasets:

```
011_011 SIC
-----------------------------------------------------------------------
netcdf ice_conc_mosaic_baltic_20250216155158_20250218050525 {
dimensions:
      time = 1 ;
      latitude = 2991 ;
      longitude = 2530 ;
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
      float latitude(latitude) ;
            latitude:long_name = "latitude" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      short ice_concentration(time, latitude, longitude) ;
            ice_concentration:long_name = "FMI ice chart sea ice con-
centration in percents" ;
            ice_concentration:standard_name = "sea_ice_area_fraction"
;
            ice_concentration:units = "%" ;
            ice_concentration:coordinates = "longitude latitude" ;
            ice_concentration:grid_mapping = "crs" ;
            ice_concentration:source = "fmi.fi" ;
            ice_concentration:_FillValue = -99s ;
      short ice_concentration_uncertainty(time, latitude, longitude) ;
            ice_concentration_uncertainty:long_name = "sea ice concen-
tration uncertainty" ;
            ice_concentration_uncertainty:units = "%" ;
            ice_concentration_uncertainty:coordinates = "longitude la-
titude" ;
            ice_concentration_uncertainty:grid_mapping = "crs" ;
            ice_concentration_uncertainty:source = "fmi.fi" ;
            ice_concentration_uncertainty:_FillValue = -99s ;

// global attributes:
            :title = "Baltic Ice Concentration Mosaic, L4, 0.5km daily
(FMI-BAL-SEAICE_CONC_MOSAIC-SAR-NRT-OBS)" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice Concentration" ;
            :product_version = "1.0" ;
```

```
            :abstract = "Baltic Sea Ice Concentration (IT) mosaic
based on SAR and MWR data. Using AMSR2, SENTINEL-1 and RADARSAT-2 im-
agery
 during the period 20250216155158 20250218050525" ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-16T15:51:58Z" ;
            :stop_date = "2025-02-18T05:05:25Z" ;
            :creation_date = "2025-02-18T09:37:31Z" ;
            :produced_date = "2025-02-18T09:37:31Z" ;
            :valid_date = "2025-02-18T05:00:00Z" ;
}

----------------------------------------------------------------------
011_011 C-band SIT (wide swath)
----------------------------------------------------------------------
netcdf ice_thickness_baltic_20250218050419 {
dimensions:
      time = 1 ;
      latitude = 1308 ;
      longitude = 1361 ;
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
      float latitude(lattitude) ;
            latitude:long_name = "latitude" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      float sea_ice_thickness(time, lat, lon) ;
            sea_ice_thickness:long_name = "SAR based sea ice thick-
ness" ;
            sea_ice_thickness:standard_name = "sea_ice_thickness" ;
            sea_ice_thickness:units = "m" ;
            sea_ice_thickness:coordinates = "longitude latitude" ;
            sea_ice_thickness:grid_mapping = "crs" ;
```

```
            sea_ice_thickness:source = "fmi.fi" ;
            sea_ice_thickness:_FillValue = -99.f ;
            sea_ice_thickness:scale_factor = 1.f ;
            sea_ice_thickness:add_offset = 0.f ;

// global attributes:
            :title = "Baltic Ice Thickness, L4 SAR, 0.5km daily (FMI-
BAL-SEAICE_THICK-SAR-NRT-OBS)" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice Thickness" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice Thickness (IT) based on IT
history from earlier IC and SAR data." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-18T05:04:19Z" ;
            :stop_date = "2025-02-18T05:04:19Z" ;
            :creation_date = "2025-02-18T05:57:48Z" ;
            :produced_date = "2025-02-18T05:57:48Z" ;
            :valid_date = "2025-02-18T05:04:19Z" ;
}

----------------------------------------------------------------------
011_011 C-band SIT (S-1 IW)
----------------------------------------------------------------------
netcdf ice_thickness_baltic_IW_20250218050027 {
dimensions:
      time = 1 ;
      latitude = 343 ;
      longitude = 471 ;
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
      float latitude(latitude) ;
            latitude:long_name = "latitude" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
                                    crs:false_northing = 0.f ;
```

```
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      float sea_ice_thickness(time, latitude, longitude) ;
            sea_ice_thickness:long_name = "SAR (S-1) based sea ice
thickness" ;
            sea_ice_thickness:standard_name = "sea_ice_thickness" ;
            sea_ice_thickness:units = "m" ;
            sea_ice_thickness:coordinates = "longitude latitude" ;
            sea_ice_thickness:grid_mapping = "crs" ;
            sea_ice_thickness:source = "fmi.fi" ;
            sea_ice_thickness:_FillValue = -99.f ;

// global attributes:
            :title = "Baltic Ice Thickness, L4 SAR, 0.5km daily (FMI-
BAL-SEAICE_THICK-SAR-IW-NRT-OBS)" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice Thickness" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice Thickness (IT) based on IT
history from earlier IC and S-1 SAR data." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-18T05:00:27Z" ;
            :stop_date = "2025-02-18T05:00:27Z" ;
            :creation_date = "2025-02-18T07:06:12Z" ;
            :produced_date = "2025-02-18T07:06:12Z" ;
            :valid_date = "2025-02-18T05:00:27Z" ;
}

----------------------------------------------------------------------
011_011 X-band SIT
----------------------------------------------------------------------
netcdf ice_thickness_baltic_X_20250218044010 {
dimensions:
      time = 1 ;
      latitude = 496 ;
      longitude = 609 ;
variables:
      int time(time) ;
            time:axis = "T" ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
                                    latitude:long_name = "latitude" ;
```

```
            latitude:axis = "Y" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:standard_name = "longitude" ;
            longitude:axis = "X" ;
            longitude:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      float sea_ice_thickness(time, latitude, longitude) ;
            sea_ice_thickness:long_name = "SAR (CSK/TSX) based sea ice
thickness" ;
            sea_ice_thickness:standard_name = "sea_ice_thickness" ;
            sea_ice_thickness:units = "m" ;
            sea_ice_thickness:coordinates = "latitude longitude" ;
            sea_ice_thickness:grid_mapping = "crs" ;
            sea_ice_thickness:source = "fmi.fi" ;
            sea_ice_thickness:_FillValue = -99.f ;

// global attributes:
            :title = "Baltic Ice Thickness, L4 SAR, 0.5km" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice Thickness" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice Thickness (IT) based on IT
history from earlier IC and X-band SAR (CSK,TSX) data." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-18T04:40:10Z" ;
            :stop_date = "2025-02-18T04:40:10Z" ;
            :creation_date = "2025-02-18T08:18:38Z" ;
            :produced_date = "2025-02-18T08:18:38Z" ;
            :valid_date = "2025-02-18T04:40:10Z" ;
}

----------------------------------------------------------------------
011_011 SIT mosaic
----------------------------------------------------------------------
netcdf ice_thick_mosaic_baltic_20250213044921_20250218055545 {
dimensions:
      time = 1 ;
      latitude = 2990 ;
                                 longitude = 2530 ;
```

```
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
      float latitude(latitude) ;
            latitude:long_name = "latitude" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      short ice_thickness(time, latitude, longitude) ;
            ice_thickness:long_name = "Baltic sea ice thickness based
on SAR data in cm" ;
            ice_thickness:standard_name = "sea_ice_thickness" ;
            ice_thickness:units = "cm" ;
            ice_thickness:coordinates = "longitude latitude" ;
            ice_thickness:grid_mapping = "crs" ;
            ice_thickness:source = "fmi.fi" ;
            ice_thickness:_FillValue = -99s ;
      short data_age(time, lat, lon) ;
            data_age:long_name = "SAR data age in hours from last ac-
quisition" ;
            data_age:units = "h" ;
            data_age:coordinates = "longitude latitude" ;
            data_age:grid_mapping = "crs" ;
            data_age:source = "fmi.fi" ;
            data_age:_FillValue = -99s ;

// global attributes:
            :title = "Baltic Ice Thickness Mosaic, L4, 0.5km daily" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice Thickness" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice Thickness (IT) mosaic based on
C- and X-band SAR imagery during the period 20250213044921 202502180
55545" ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
```

```
            :start_date = "2025-02-13T04:49:21Z" ;
            :stop_date = "2025-02-18T05:55:45Z" ;
            :creation_date = "2025-02-18T08:02:25Z" ;
            :produced_date = "2025-02-18T08:02:25Z" ;
            :valid_date = "2025-02-18T08:00:00Z" ;
}

----------------------------------------------------------------------
011_011 SID
----------------------------------------------------------------------
netcdf ice_drift_baltic_20250217044145_20250218045018 {
dimensions:
      time = 1 ;
      nv = 2 ;
      latitude = 366 ;
      longitude = 429 ;
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:bounds = "time_bnds" ;
            time:calendar = "standard" ;
      int time_bnds(time, nv) ;
            time_bnds:units = "seconds since 1981-01-01 00:00:00" ;
      float latitude(latitude) ;
            latitude:long_name = "latitude" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      float sea_ice_x_displacement(time, latitude, longitude) ;
            sea_ice_x_displacement:long_name = "sea ice drift U coor-
dinate" ;
            sea_ice_x_displacement:standard_name =
"sea_ice_x_displacement" ;
            sea_ice_x_displacement:units = "m" ;
            sea_ice_x_displacement:coordinates = "longitude latitude"
;
            sea_ice_x_displacement:grid_mapping = "crs" ;
            sea_ice_x_displacement:source = "fmi.fi" ;
            sea_ice_x_displacement:_FillValue = -99999.f ;
            sea_ice_x_displacement:scale_factor = 1.f ;
            sea_ice_x_displacement:add_offset = 0.f ;
      float sea_ice_y_displacement(time, lat, lon) ;
            sea_ice_y_displacement:long_name = "sea ice drift V coor-
dinate" ;
            sea_ice_y_displacement:standard_name =
"sea_ice_y_displacement" ;
```

```
            sea_ice_y_displacement:units = "m" ;
            sea_ice_y_displacement:coordinates = "longitude latitude"
;
            sea_ice_y_displacement:grid_mapping = "crs" ;
            sea_ice_y_displacement:source = "fmi.fi" ;
            sea_ice_y_displacement:_FillValue = -99999.f ;
            sea_ice_y_displacement:scale_factor = 1.f ;
            sea_ice_y_displacement:add_offset = 0.f ;
      short ice_drift_quality(time, lat, lon) ;
            ice_drift_quality:long_name = "sea ice drift quality" ;
            ice_drift_quality:units = "%" ;
            ice_drift_quality:coordinates = "lon lat" ;
            ice_drift_quality:grid_mapping = "crs" ;
            ice_drift_quality:source = "fmi.fi" ;

// global attributes:
            :title = "Baltic Ice Drift, L4 SAR, 0.8km daily (FMI-BAL-
SEAICE_DRIFT-SAR-NRT-OBS)" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice drift" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice drift based on two SAR images
acquired at different time instants." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-17T04:41:45Z" ;
            :stop_date = "2025-02-18T04:50:18Z" ;
            :creation_date = "2025-02-18T09:17:52Z" ;
            :produced_date = "2025-02-18T09:17:52Z" ;
            :valid_date = "2025-02-18T04:50:18Z" ;
}

----------------------------------------------------------------------
011_011 SIE
----------------------------------------------------------------------
netcdf sie_baltic_eo_20250217 {
dimensions:
      time = UNLIMITED ; // (1 currently)
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:calendar = "standard" ;
      float sea_ice_extent(time) ;
```

```
            sea_ice_extent:long_name = "sea ice extent" ;
            sea_ice_extent:standard_name = "sea_ice_extent" ;
            sea_ice_extent:units = "1e6km2" ;

// global attributes:
            :title = "Baltic Sea Ice Extent (EO), L4, 1km daily" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice extent" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice Extent (SIE) based on EO da-
ta." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 906.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-17T14:00:00Z" ;
            :stop_date = "2025-02-17T14:00:00Z" ;
            :creation_date = "2025-02-17T15:45:02Z" ;
            :produced_date = "2025-02-17T15:45:02Z" ;
            :valid_date = "2025-02-17T14:00:00Z" ;
}

-----------------------------------------------------------------------
011_011 SID mosaic
-----------------------------------------------------------------------
netcdf ice_drift_mosaic_baltic_20250225140000 {
dimensions:
      time = 1 ;
      nv = 2 ;
      latitude = 1857 ;
      longitude = 1467 ;
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:bounds = "time_bnds" ;
            time:calendar = "standard" ;
      int time_bnds(time, nv) ;
            time_bnds:units = "seconds since 1981-01-01 00:00:00" ;
      float latitude(latitude) ;
            latitude:long_name = "latitude" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:units = "degrees_east" ;
      char crs ;
```

```
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      float sea_ice_x_velocity(time, latitude, longitude) ;
            sea_ice_x_velocity:long_name = "sea ice drift velocity_U
coordinate" ;
            sea_ice_x_velocity:standard_name = "sea_ice_x_velocity" ;
            sea_ice_x_velocity:units = "mm/s" ;
            sea_ice_x_velocity:coordinates = "longitude latitude" ;
            sea_ice_x_velocity:grid_mapping = "crs" ;
            sea_ice_x_velocity:source = "fmi.fi" ;
            sea_ice_x_velocity:_FillValue = -99999.f ;
      float sea_ice_y_velocity(time, lat, lon) ;
            sea_ice_y_velocity:long_name = "sea ice drift_velocity V
coordinate" ;
            sea_ice_y_velocity:standard_name = "sea_ice_y_velocity" ;
            sea_ice_y_velocity:units = "mm/s" ;
            sea_ice_y_velocity:coordinates = "lon lat" ;
            sea_ice_y_velocity:grid_mapping = "crs" ;
            sea_ice_y_velocity:source = "fmi.fi" ;
            sea_ice_y_velocity:_FillValue = -99999.f ;
      float sea_ice_drift_quality(time, lat, lon) ;
            sea_ice_drift_quality:long_name = "sea ice drift quality"
;
            sea_ice_dnetcdf ice_drift_mosaic_baltic_20250225140000 {
dimensions:
      time = 1 ;
      nv = 2 ;
      latitude = 1857 ;
      longitude = 1467 ;
variables:
      int time(time) ;
            time:long_name = "reference time of the product" ;
            time:standard_name = "time" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:bounds = "time_bnds" ;
            time:calendar = "standard" ;
      int time_bnds(time, nv) ;
            time_bnds:units = "seconds since 1981-01-01 00:00:00" ;
      float latitude(latitude) ;
            latitude:long_name = "latitude" ;
            latitude:units = "degrees_north" ;
      float longitude(longitude) ;
            longitude:long_name = "longitude" ;
            longitude:units = "degrees_east" ;
      char crs ;
            crs:grid_mapping_name = "latitude_longitude" ;
            crs:false_easting = 0.f ;
            crs:false_northing = 0.f ;
            crs:proj4_string = "proj=latlong datum=WGS84" ;
      float sea_ice_x_velocity(time, latitude, longitude) ;
            sea_ice_x_velocity:long_name = "sea ice drift velocity_U
coordinate" ;
```

```
            sea_ice_x_velocity:standard_name = "sea_ice_x_velocity" ;
            sea_ice_x_velocity:units = "mm/s" ;
            sea_ice_x_velocity:coordinates = "longitude latitude" ;
            sea_ice_x_velocity:grid_mapping = "crs" ;
            sea_ice_x_velocity:source = "fmi.fi" ;
            sea_ice_x_velocity:_FillValue = -99999.f ;
      float sea_ice_y_velocity(time, latitude, longitude) ;
            sea_ice_y_velocity:long_name = "sea ice drift_velocity V
coordinate" ;
            sea_ice_y_velocity:standard_name = "sea_ice_y_velocity" ;
            sea_ice_y_velocity:units = "mm/s" ;
            sea_ice_y_velocity:coordinates = "longitude latitude" ;
            sea_ice_y_velocity:grid_mapping = "crs" ;
            sea_ice_y_velocity:source = "fmi.fi" ;
            sea_ice_y_velocity:_FillValue = -99999.f ;
      float sea_ice_drift_quality(time, latitude, longitude) ;
            sea_ice_drift_quality:long_name = "sea ice drift quality"
;
            sea_ice_drift_quality:units = "%" ;
            sea_ice_drift_quality:coordinates = "lon lat" ;
            sea_ice_drift_quality:_FillValue = -99999.f ;
            sea_ice_drift_quality:grid_mapping = "crs" ;
            sea_ice_drift_quality:source = "fmi.fi" ;

// global attributes:
            :title = "Baltic Ice Drift mosaic, L4 SAR, 0.8km daily
(cmems_obs-si_bal_phy-sid_nrt_l4_P1D-m)" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice drift mosaic" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice drift velocity mosaic based on
two SAR images pairs." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-22T16:21:38Z" ;
            :stop_date = "2025-02-25T16:14:08Z" ;
            :creation_date = "2025-02-27T14:38:37Z" ;
            :produced_date = "2025-02-27T14:38:37Z" ;
            :valid_date = "2025-02-25T14:00:00Z" ;rift_quality:units =
"%" ;
            sea_ice_drift_quality:coordinates = "lon lat" ;
            sea_ice_drift_quality:_FillValue = -99999.f ;
            sea_ice_drift_quality:grid_mapping = "crs" ;
            sea_ice_drift_quality:source = "fmi.fi" ;

                        // global attributes:
```

```
            :title = "Baltic Ice Drift mosaic, L4 SAR, 0.8km daily
(cmems_obs-si_bal_phy-sid_nrt_l4_P1D-m)" ;
            :Conventions = "CF-1.4" ;
            :product_name = "Baltic Sea Ice drift mosaic" ;
            :product_version = "1.0" ;
            :abstract = "Baltic Sea Ice drift velocity mosaic based on
two SAR images pairs." ;
            :area = "Baltic Sea" ;
            :institution = "Finnish Meteorological Institute (FMI)" ;
            :contact = "juha.karvonen@fmi.fi" ;
            :operational_status = "operational" ;
            :references = "" ;
            :history = "" ;
            :netcdf_version_id = "4" ;
            :southernmost_latitude = -90.f ;
            :northernmost_latitude = 90.f ;
            :westernmost_longitude = -180.f ;
            :easternmost_longitude = 180.f ;
            :start_date = "2025-02-22T16:21:38Z" ;
            :stop_date = "2025-02-25T16:14:08Z" ;
            :creation_date = "2025-02-27T14:38:37Z" ;
            :produced_date = "2025-02-27T14:38:37Z" ;
            :valid_date = "2025-02-25T14:00:00Z" ;
}
```